

Audio Forensics Analysis

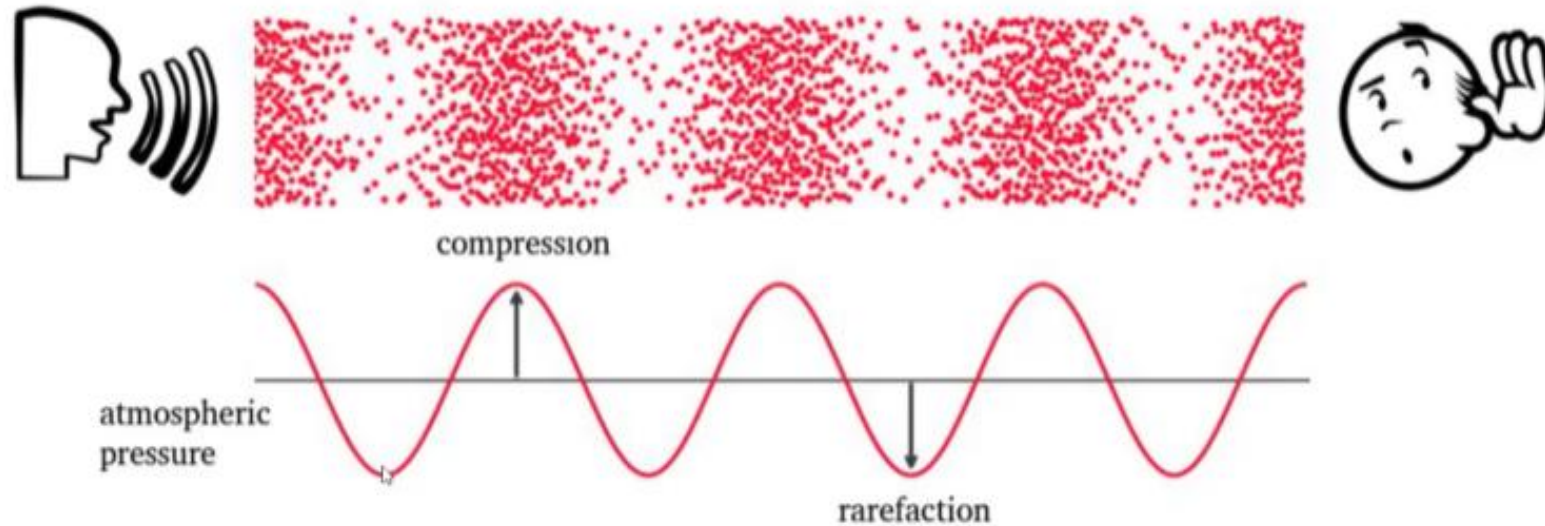
National Forensic Sciences University, Dharwad Campus

Training on Audio Forensics

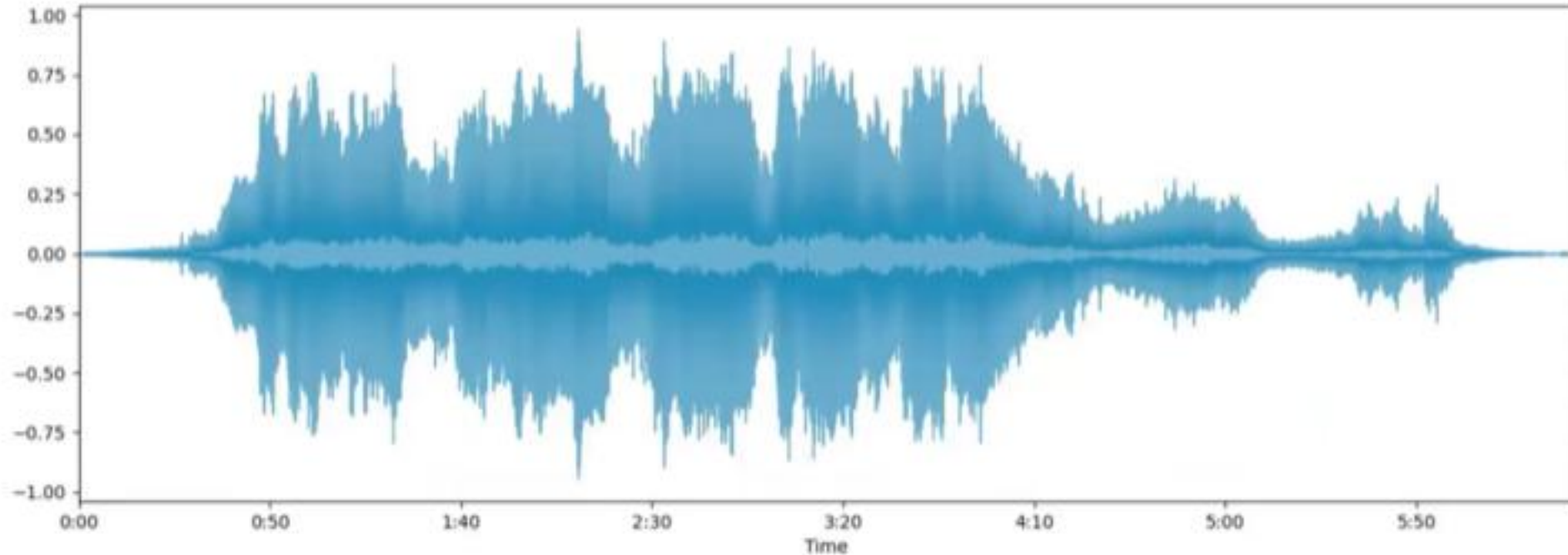
Date: 8th November 2025

Sound Production and Wave representation

Sound wave

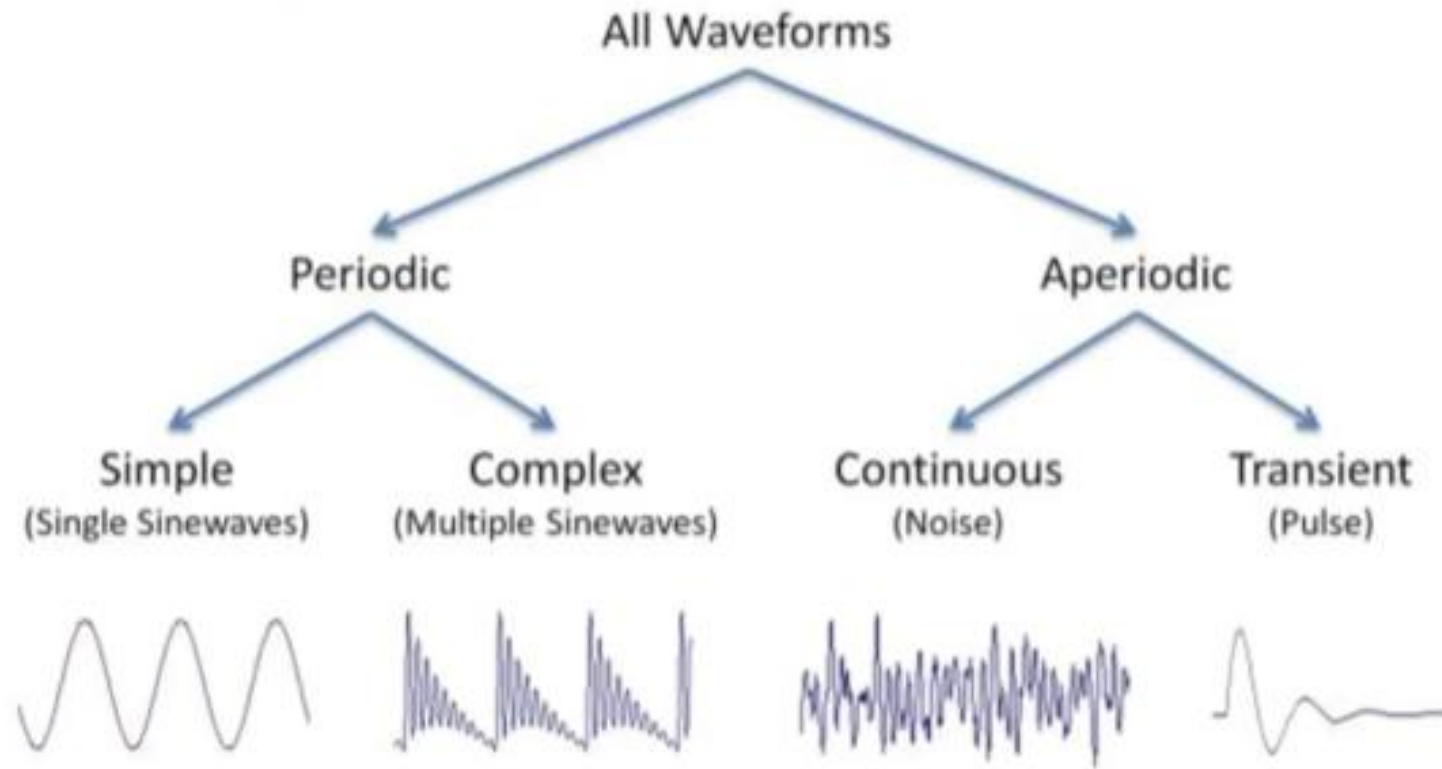


Waveform



- **Visualizing Sound:** Waveforms can be seen with tools like **Audacity**, showing amplitude/pressure (y-axis) versus time (x-axis).
- **Multifactorial information of waveform:** Frequency, Intensity, pitch, etc

Periodic and aperiodic sound

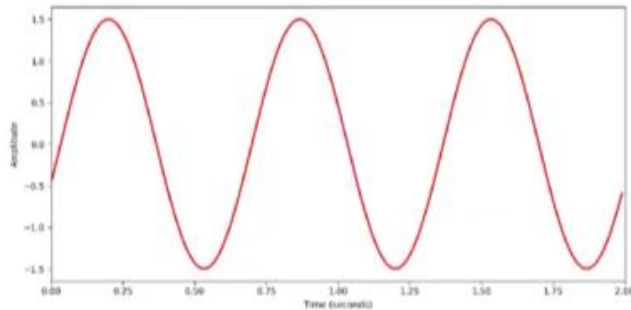


Key Parameters of Audio Signals

- **Frequency (Hz):** Number of oscillations per second (cycle/sec). Inverse of the time period.
Defines *pitch*—higher frequencies = higher pitches.
 - Human hearing: 20 Hz to 20,000 Hz (20 kHz).
 - Male speech: ~60–5,500 Hz. Female speech: ~150–10,000 Hz.
 - Animals: Cats up to 75,000 Hz, dogs up to 45,000 Hz, bats up to 200,000 Hz.
- **Amplitude:** The maximum displacement of the wave from reference. Higher amplitude = louder sound. Amplitude measured from the middle (zero pressure) to peak pressure.

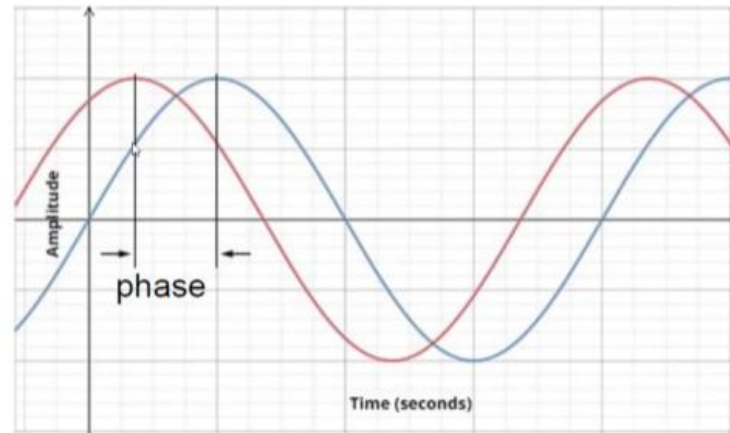
Key Parameters of Audio Signals

Waveform

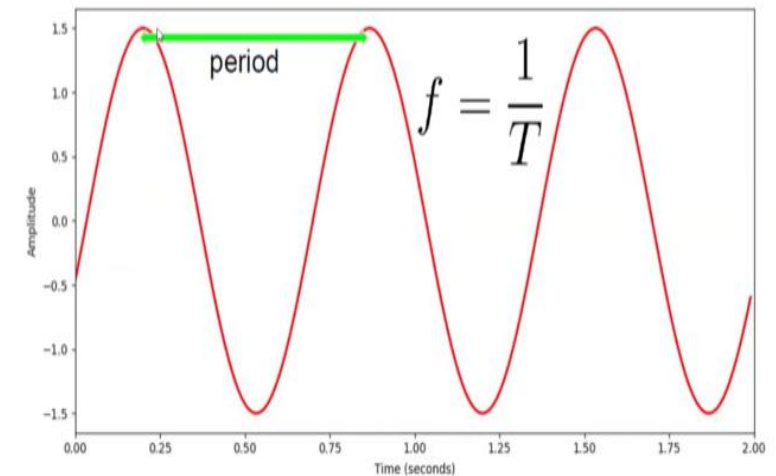


$$y(t) = A \sin(2\pi ft + \varphi)$$

Phase



Frequency



Key Parameters of Audio Signals

- **Phase:** The relative shift between two waves, measured in degrees or radians. Important in wave superposition—two waves of the same amplitude and opposite phase can cancel each other out (destructive interference).
- **Intensity (W/m^2):** Power transported by the wave per unit area. Human hearing threshold: 10^{-12} W/m^2 . Threshold of pain: 10 W/m^2 .
- **Loudness (Phons):** Human-perceived strength of a sound, a subjective correlate of intensity. Loudness perception depends on frequency, age, and person-to-person variability.

- **Decibels (dB):** Logarithmic measurement of intensity:

$$L = 10 \log_{10}\left(\frac{I}{I_0}\right)$$

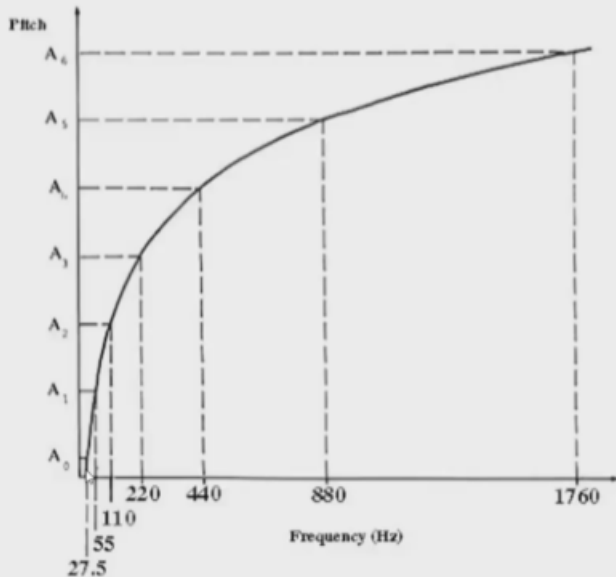
where I = measured intensity, I_0 = threshold intensity for hearing.

- Threshold of hearing: 0 dB; normal conversation: ~60 dB; painful: 130 dB; jet engine: 140 dB.

Frequency vs pitch:

- **Pitch:** Psychological perception of frequency, varies logarithmically rather than linearly.
- Pitch refers to the perception of the sound. Pitch is how we perceive with our ear.
- Frequency refers to the production of the sound.

Pitch-frequency chart



Mapping pitch to frequency

$$F(p) = 2^{\frac{p-69}{12}} \cdot 440$$

At pitch=69, F=440Hz

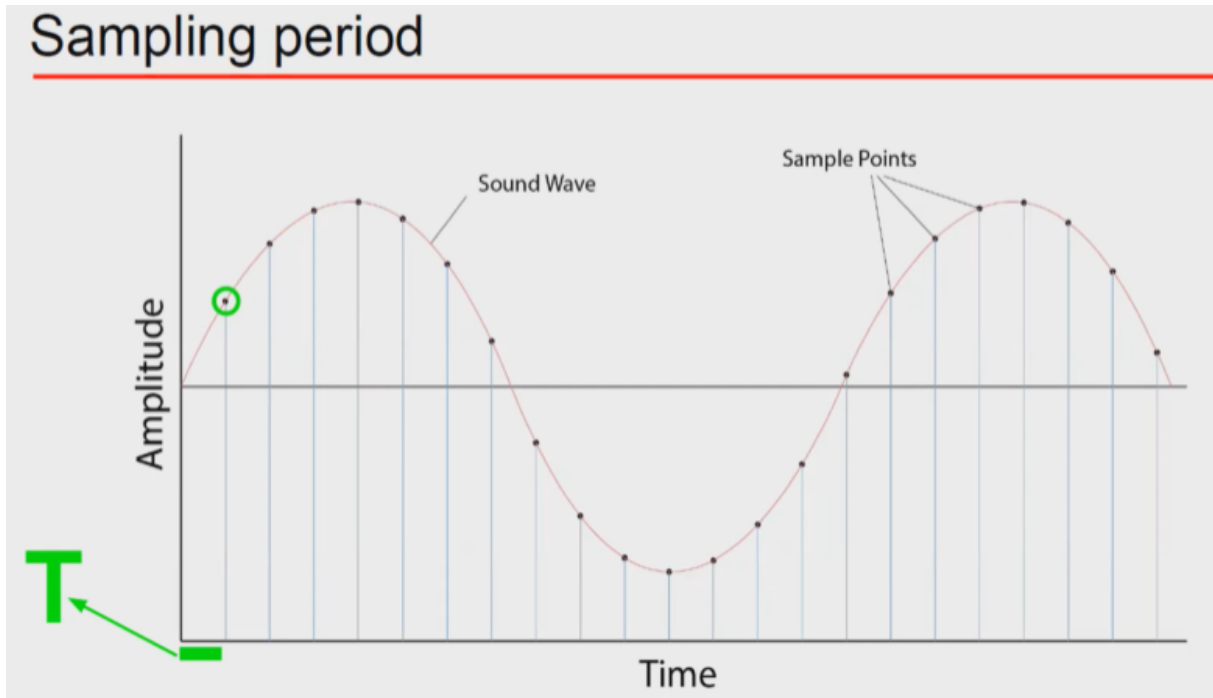
At pitch = 60, F= 261.6Hz

Mapping pitch to frequency

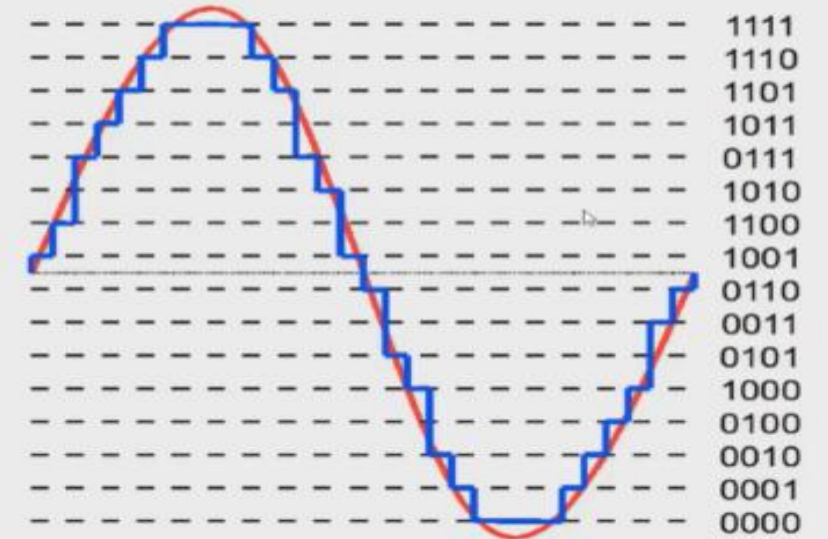
$$F(p+1)/F(p) = 2^{1/12} = 1.059$$

Analog and Digital Signals; Sampling and Quantization

Sampling period



Quantization



Sampling Frequency based on Nyquist Criteria

Aliasing is Under sampling

CDs use 44.1 kHz to cover human hearing. Max human range frequency is 20kHz. Sampling theorem:
 $F_s = 2 \times \text{Max frequency}$.

Resolution = number of bits or bit depth

CD resolution = 16 bits = 65536

Data Converters-ADC and DAC

How do we record sound?

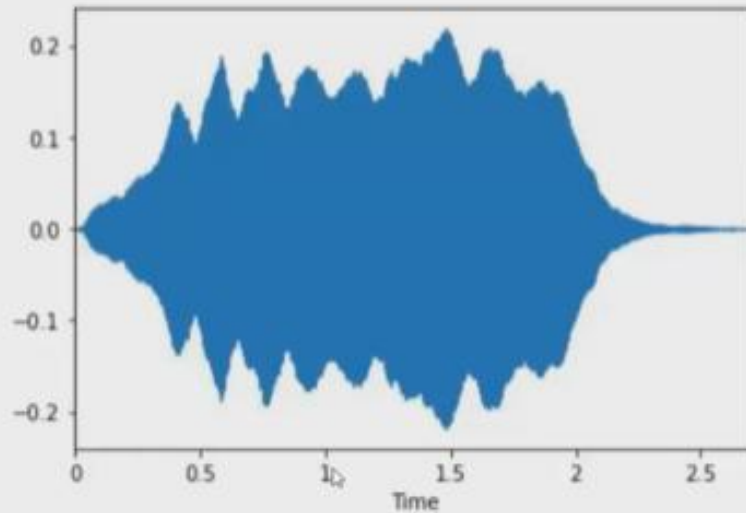


How do we reproduce sound?

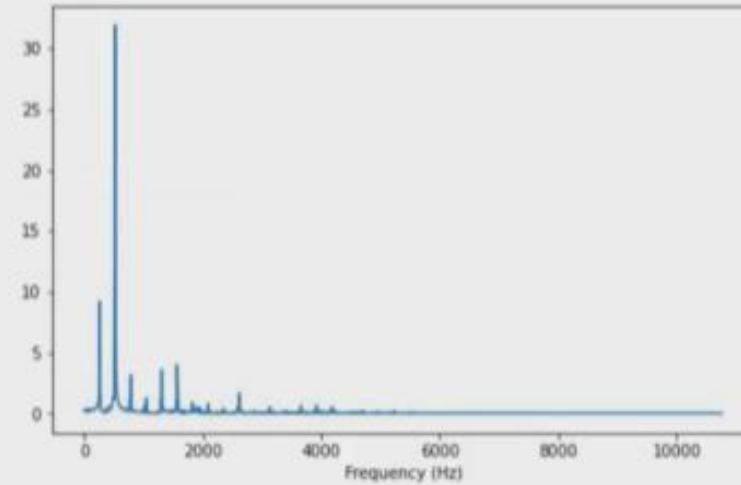


Time and Frequency Domain: Fourier

From time to frequency domain



FT



Challenges in Audio Processing and need for enhancement

- **Noise:** Unwanted sound makes extraction and analysis harder.
- **Aliasing & Quantization Errors:** Bad sampling/poor bit depth reduces fidelity.
- **Loss of Information:** Compression and downsampling can discard important details.
- **Subjectivity in Loudness/Pitch Perception:** Variability in human listeners means technical measurements don't always match perception.

Audio Authentication

- **Electronic evidence:** According to IT Act Chapter 1, sound files are included as electronic records.

Authentication Process Overview

- Lab examination
- Original/Copied
- Social Media Processed?
- Critical (Auditory) Listening

Audio Authentication

Pre-Examination Assessment

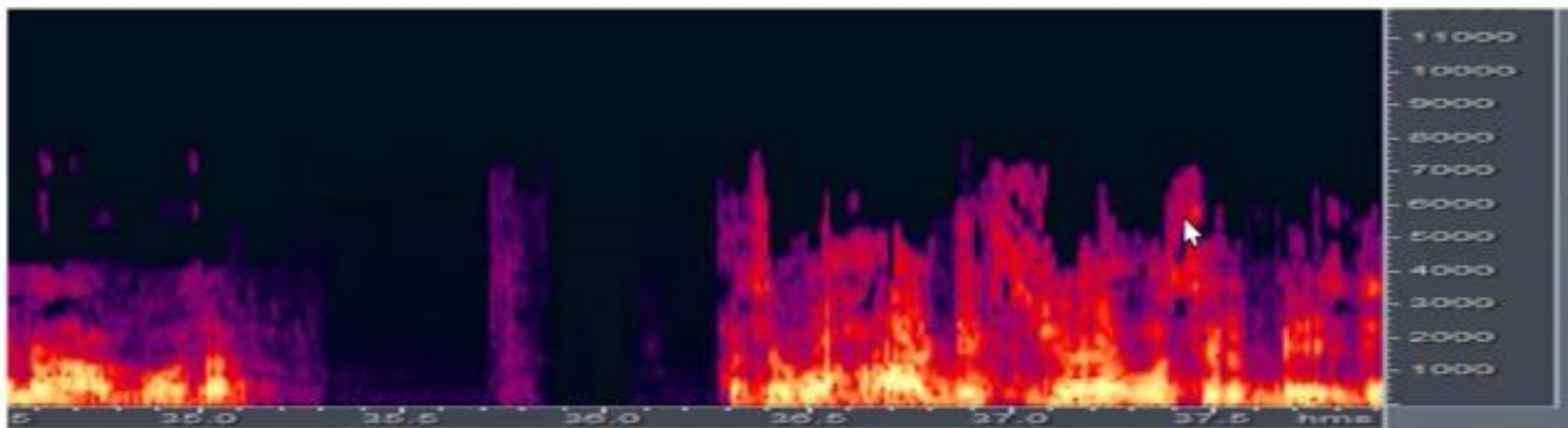
- Always request the original recording or original device.
- If unavailable, request details of recording device: make, model, serial number; date, time of copying; details of copying process.
- For telephone recordings, request Call Detail Records (CDR) for data verification like length of the recording, time, date and etc.
- Maintain and check Chain Of Custody Documents (COD).

Audio authentication-Laboratory Examination

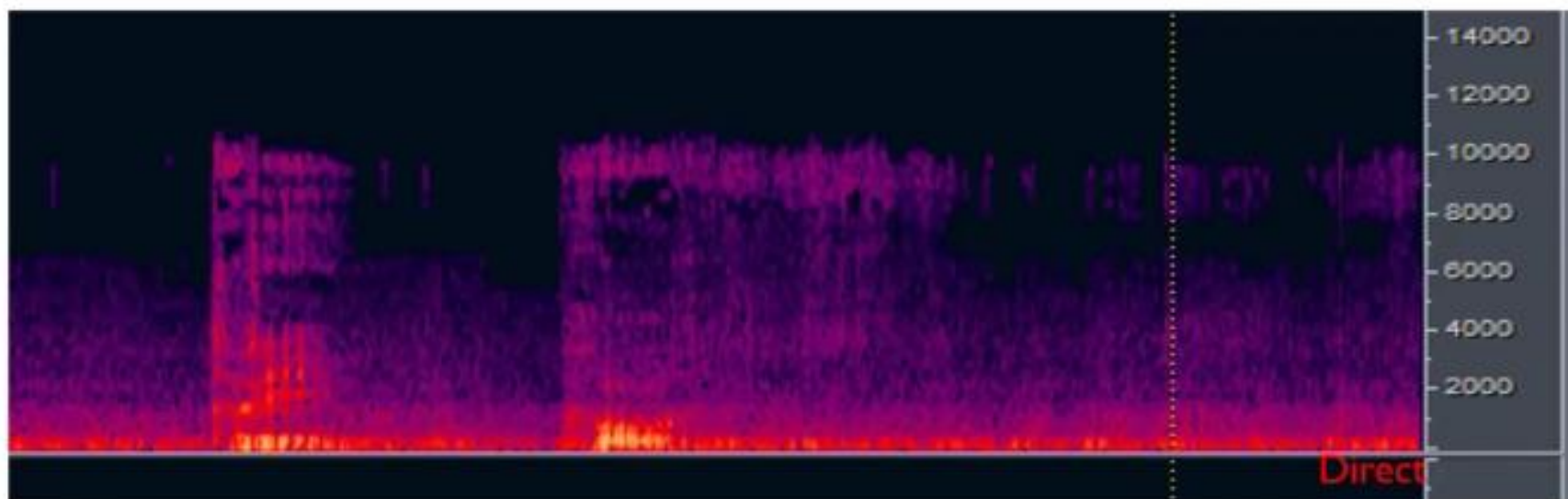
- Direct or Telephonic Recording
- Using the following features, the direct and telephonic recordings can be identified easily:
 - i. The sampling rate: 8000 Hz for Tele
23000 Hz or more for Direct
 - ii. Background Noise:

Low	Tele
High	Direct
 - iii. Metadata/HEX data properties to be verified

Direct Vs Telephonic Recording



Telephonic



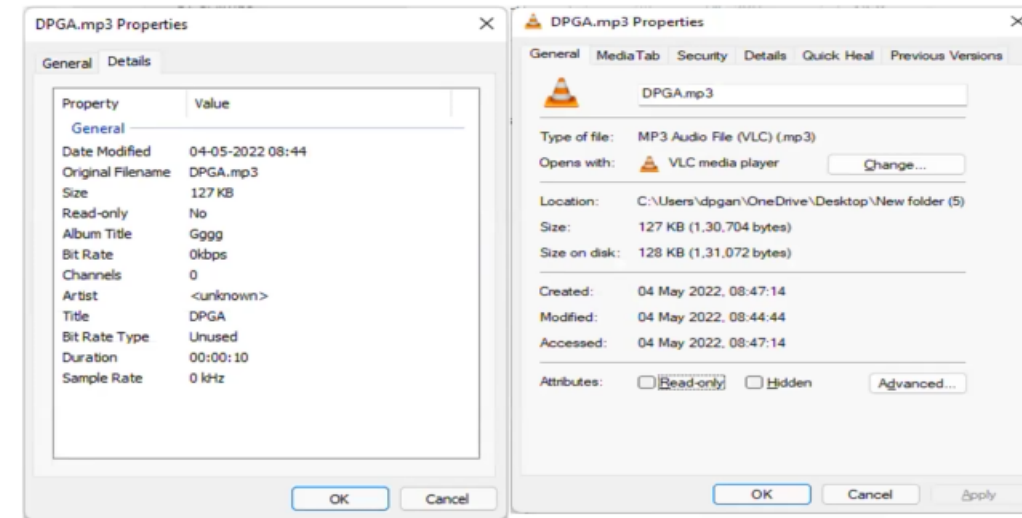
Direct

Metadata Analysis

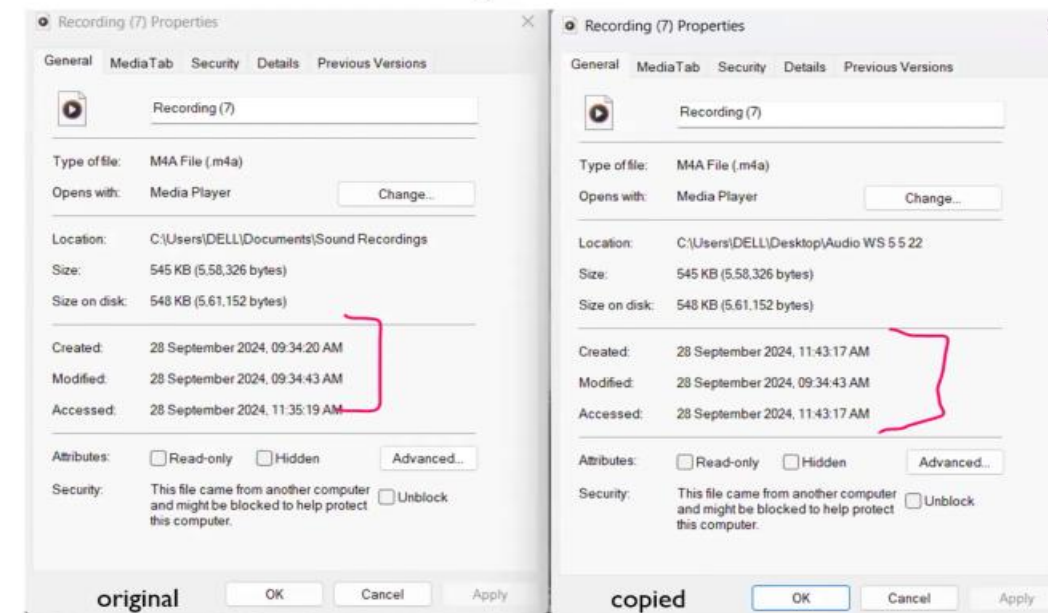
- Extracts technical info from the audio file (e.g., filename, format, duration, bitrate, encoder, hash values).
- Confirm file properties match stated characteristics
- Check hash values for integrity
- Look for signs of re-encoding or editing

2. Original/copied?

i. Using the metadata and HEX data



2. Original/copied?



Hash Values

- Inference for Audio Forensics:

If these hash values match the hashes taken from an original device, the audio file is considered authentic, untampered, and reliable as evidence.

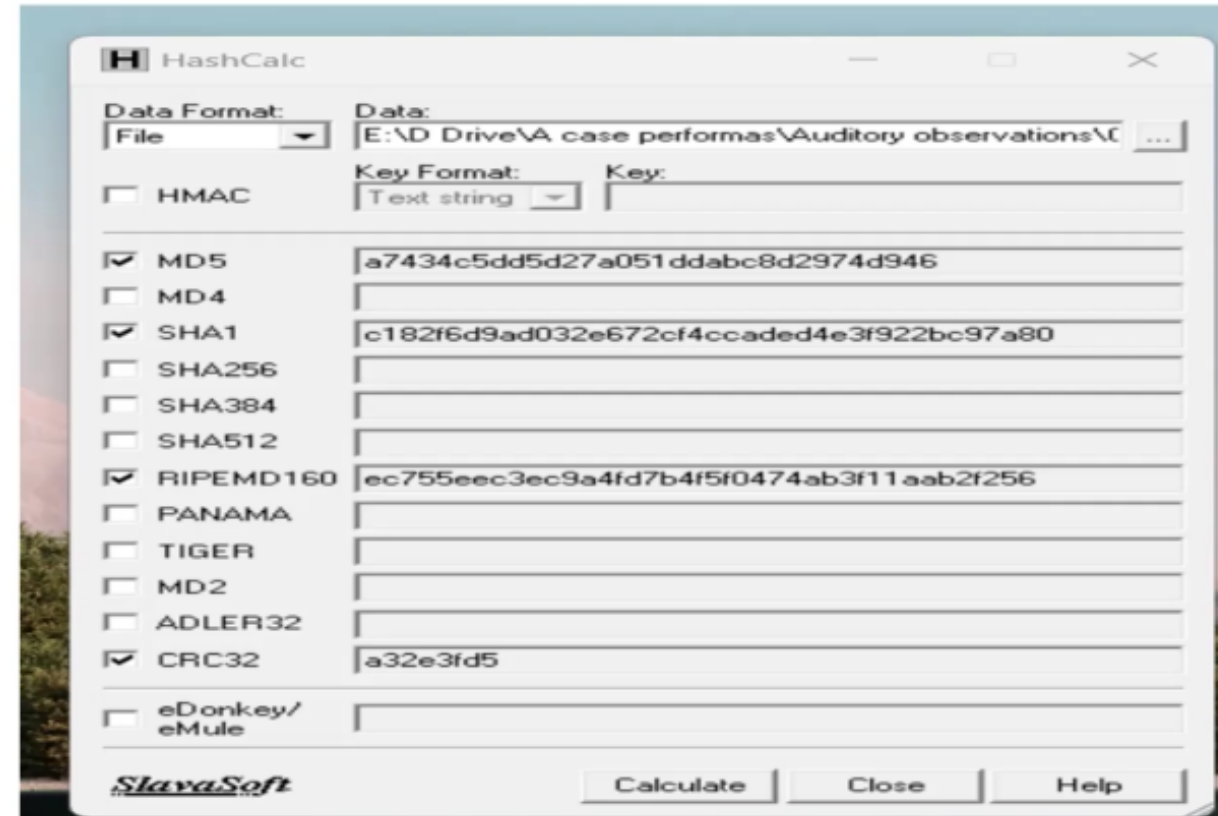
- If hashes differ, it indicates modifications, edits, or possible tampering, which must be flagged in forensic analysis and reporting.
- Analyze with OSI tools, MediaInfo, Mediatape, Hex editors, or scripts (e.g., Python).
- Check parameters: sampling rate, codec, bit rate, device name, and editing history.

- Online Hash calculator:

<https://md5file.com/calculator>

HASH VERIFICATION

Hash Cancellation: If original value is available



Waveform Analysis:

- Visual comparison to spot discontinuities, sharp cuts, or faces that don't match up.
- Difficult to detect if edits are made with technical precision (phase-aligned cutting).
- Human vs. synthesized/AI voice detection:
 - Human speech naturally exhibits variation; synthesized or programmatic voice segments show near-identical waveforms repeatedly.

1. Discontinuity

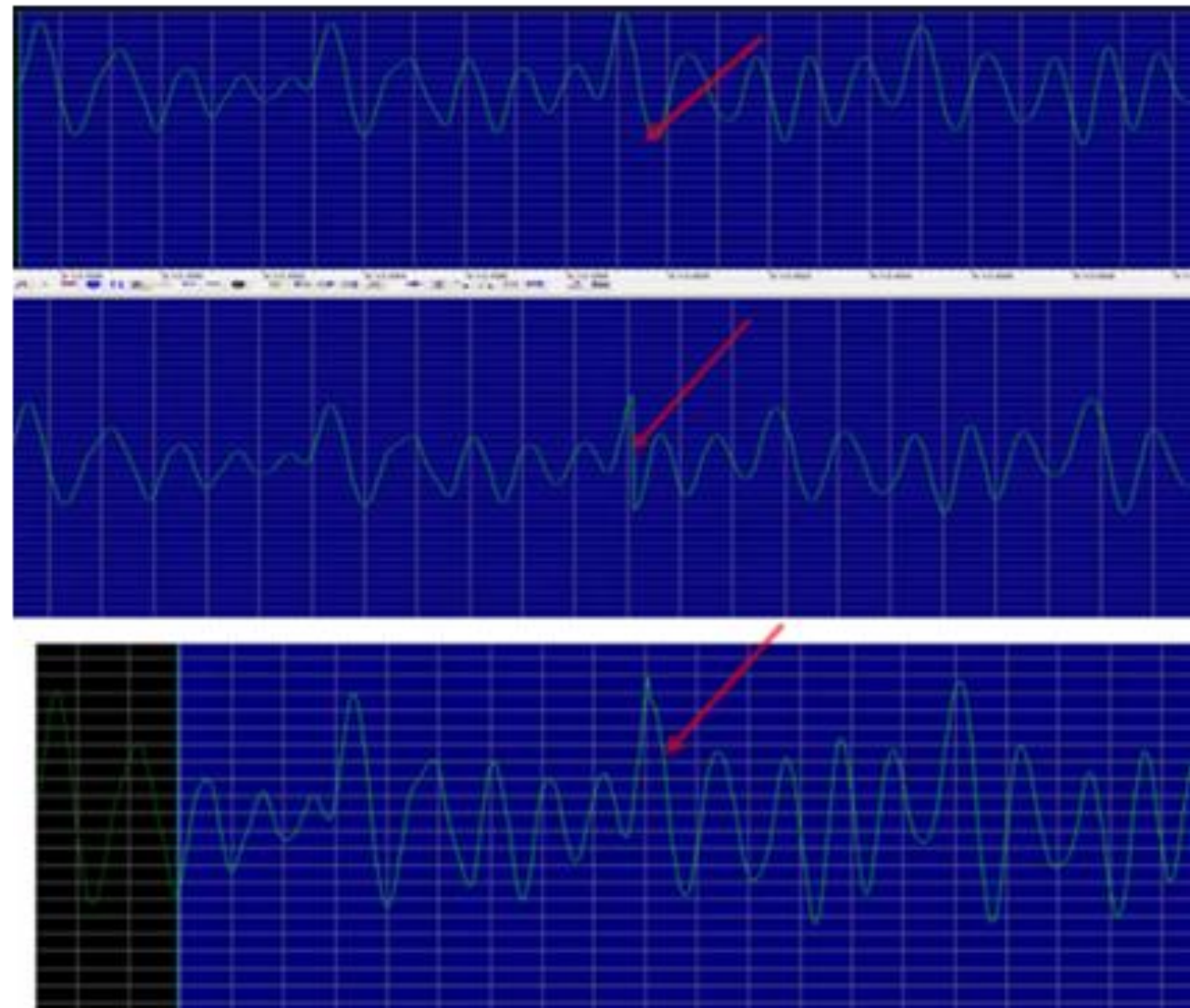
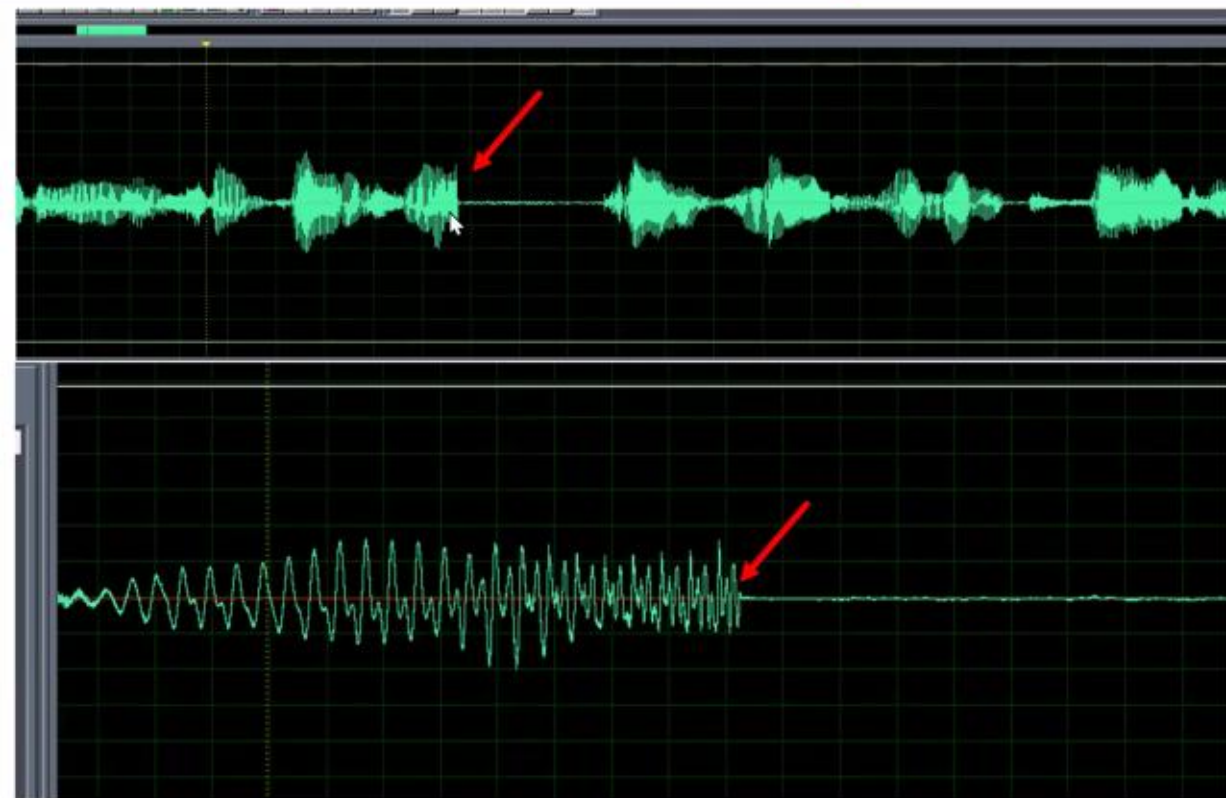
2. Human and Synthesized voice

3. Copy paste/ deletion/addition

4. Background noise variation

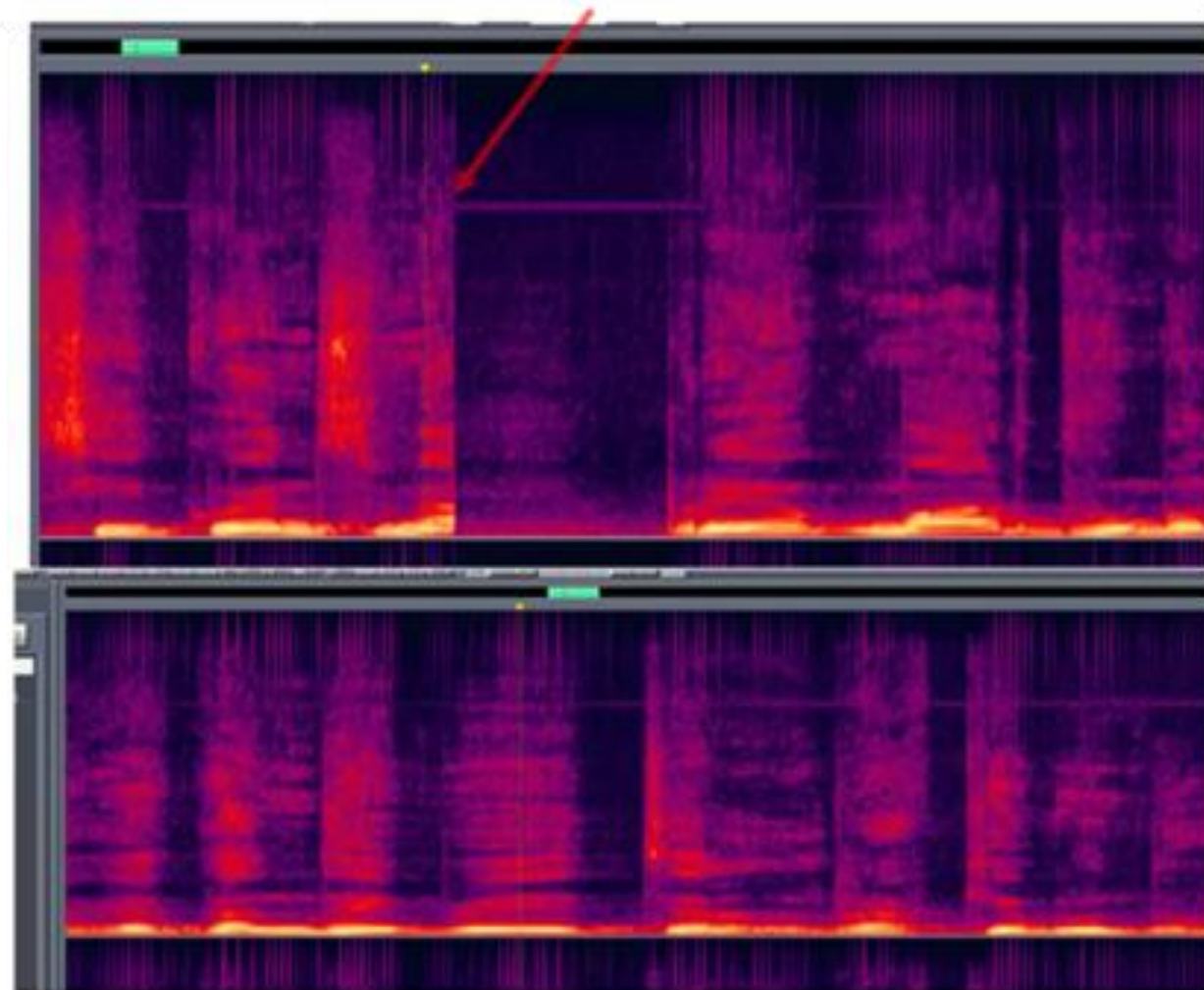
Waveform Analysis:

Wave form Analysis

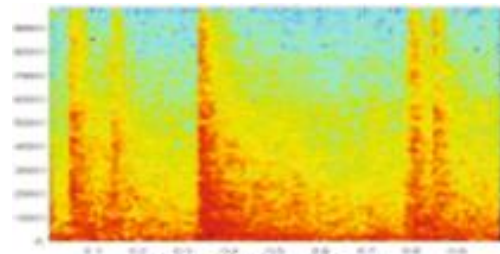


Spectro graphical analysis

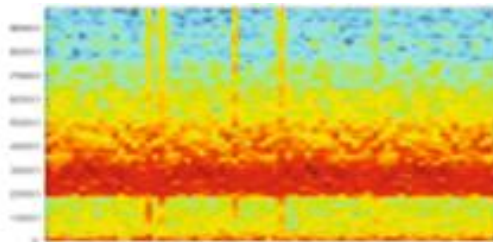
- Visualizes frequency content over time.
- Look for abrupt frequency changes or repeated patterns
- Sudden start/stop bands suggest edits or synthetic content



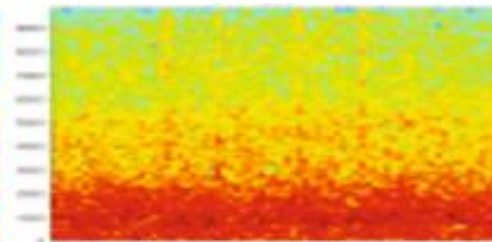
Spectrogram Analysis



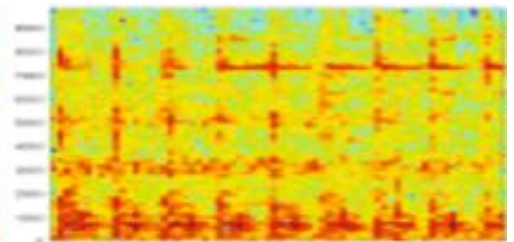
excavator



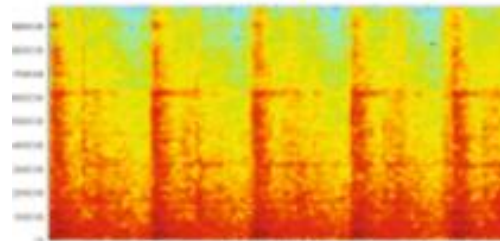
alarm



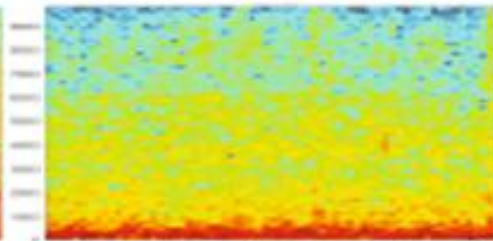
engine sound



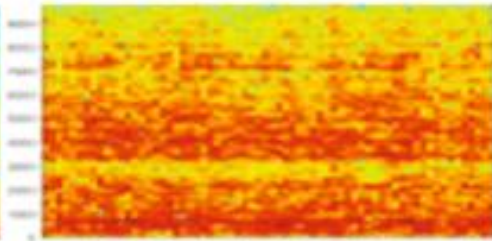
electric hammer



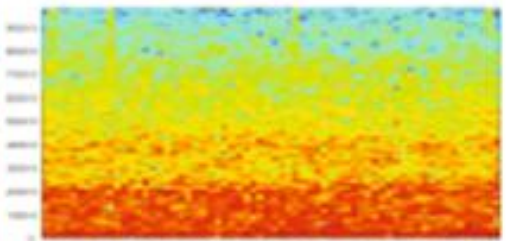
hydraulic hammer



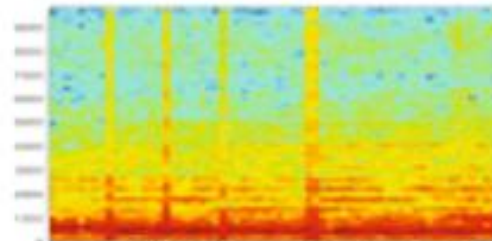
milling machine



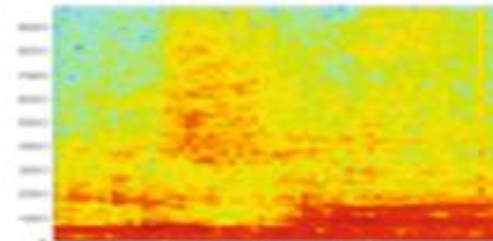
cutting machine



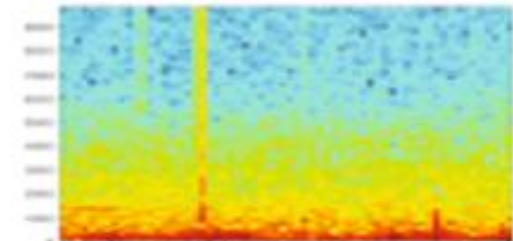
sound of generator



music



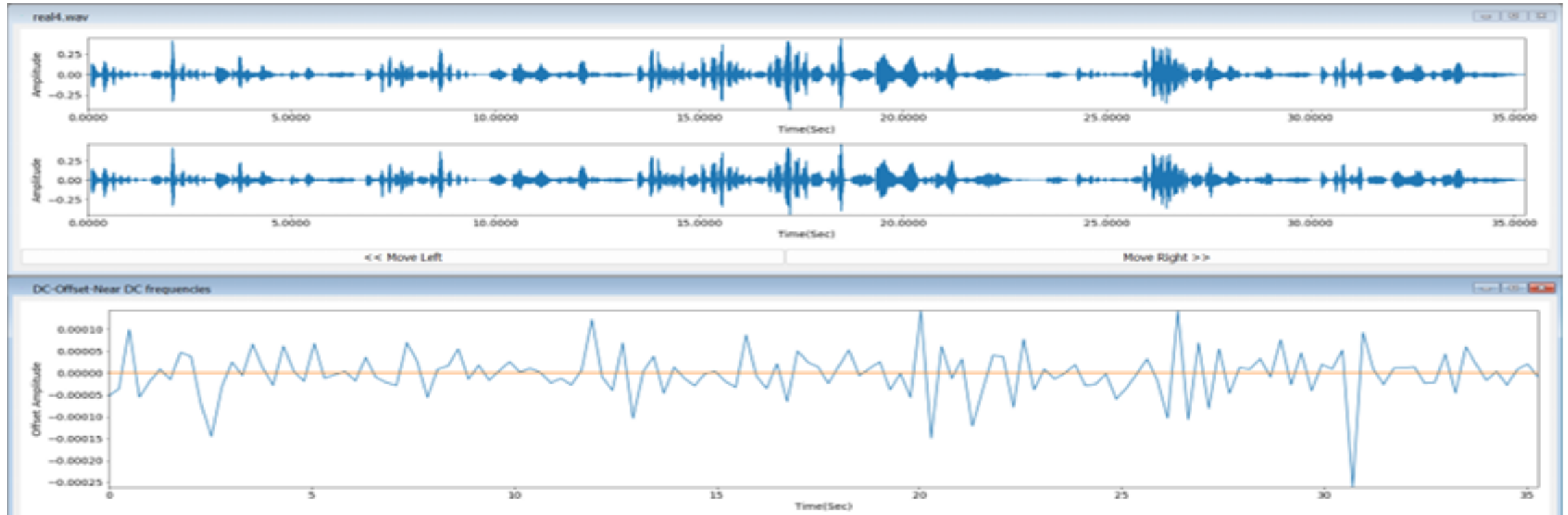
talking



wind noise

DC Shift Analysis

- Measures whether the center line of the waveform is offset from zero.
- Large DC offset = improper recording or editing



Statistical Parameters

- Skewness, Kurtosis: indicate amplitude distribution shape
- SNR (Signal-to-Noise Ratio): higher = clearer audio

Silence/Zero Discontinuity Detection

- Checks for unnatural silences or zero blocks.
- Natural speech pauses differ from hard cuts or inserted silences

LPC (Linear Predictive Coding) Discontinuity

- Analyzes prediction changes over time.
- Discontinuities = possible splicing
- 'No discontinuity found' = likely original audio

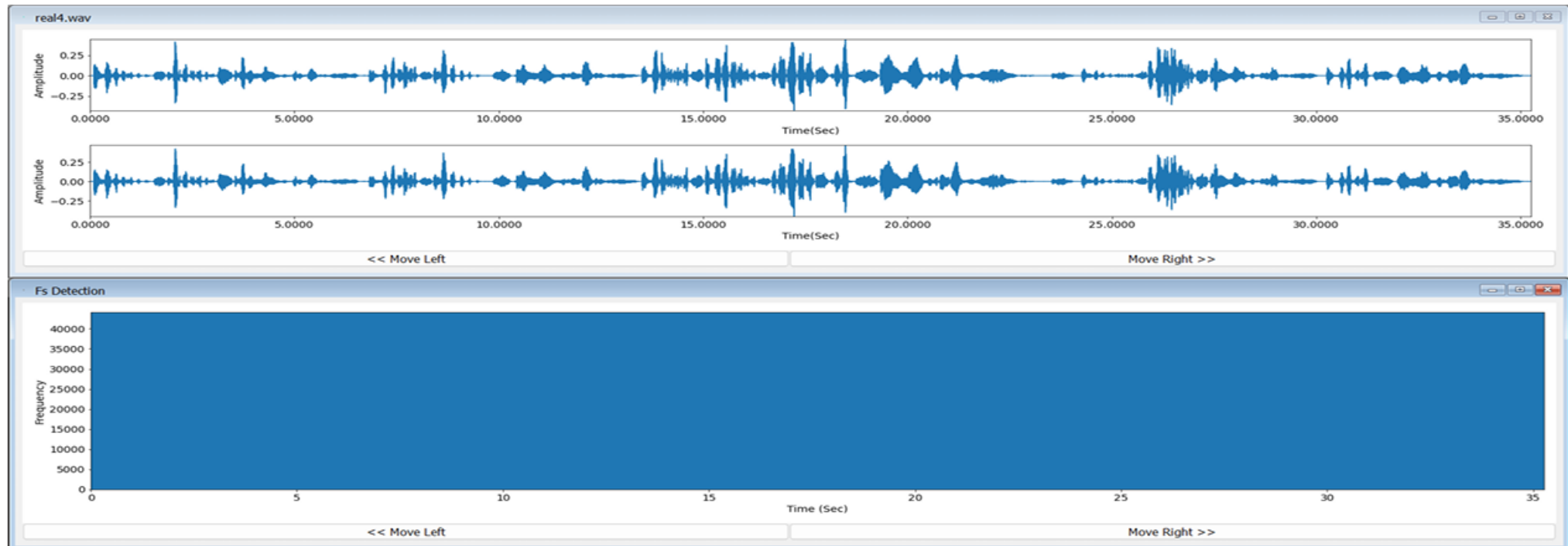
ENF (Electric Network Frequency) Variation

- Checks continuity of 50Hz power line hum.
- Absence or sudden change may suggest manipulation

Sampling Frequency Detection

- Confirms audio sampling rate (e.g., 44.1kHz).
- Mismatch may indicate resampling or re-encoding

Fs-Detection



Specific Scenarios for Authentication & Appropriate Techniques

Audio File Scenario	Techniques	Outcome/Opinion
Original file on original device like mobile phone	Metadata, Hash	Conclusive authentication possible
File copied to another device, but original available like cd or pendrive with source	Metadata, Hash, Auditory, waveform form, Spectrograph	Strong opinion possible
File only on CD/disk, not social media processed	Metadata, Auditory, waveform-form, Spectrograph	Concrete or continuity-only
File from social media but present in device	Mostly Auditory, waveform form, Spectrograph	Only continuity possible

Some OSI tools for Audio Signal Analysis

- Audacity
- Praat
- Hash Calc
- Sonic Visualizer

Thank you